66.

Solution:

Given :

$T = 175^0 C = 175 + 273 = 448K$

Concentration of 1-pentyne

[A] = 1.3%

Concentration of 2-pentyne
[B] = 95.2%

Concentration of 1, 2-pentadiene
[C] = 3.5%

At equilibrium

$B \rightleftharpoons A$

95.2% 1.3% $\Rightarrow$

$K_1 = \frac{1.3}{95.2} = 0.0136$

$B \rightleftharpoons C$

95.2% 3.5% $\Rightarrow$

$K_2 = \frac{3.5}{95.2} = 0.0367$

$\Rightarrow \Delta G_1^0 = -2.303\ RT \log K_1$

$\Delta G_1^0 = -2.303 \times 8.314 \times 448 \times \log 0.0136$

$\Delta G_1^0 = +16010\ J$

$\Delta G_1^0 = +16\ kJ$

$\Rightarrow \Delta G_2^0 = -2.303\ RT \log K_2$

$\Delta G_2^0 = -2.303 \times 8.314 \times 448 \times \log 0.0367$

$\Delta G_2^0 = +12312\ J$

$\Delta G_2^0 = +12.312\ kJ$

67.

Solution:

Given :

$T = 33K$

$N_2O_4 \rightleftharpoons 2NO_2$

	N_2O_4	$\rightleftharpoons$	$2NO_2$
Initial no of moles	100		0
Number of moles dissociated	50		-
No of moles remaining	50		100
Total no of moles	150 moles		

$P_{N_2O_4} = \frac{n_{N_2O_4}}{n_{N_2O_4} + n_{NO_2}} . P$

$P_{N_2O_4} = \frac{50\ mol}{150\ mol} .1\ atm$

$P_{N_2O_4} = 0.333\ atm$

$P_{NO_2} = \frac{n_{NO_2}}{n_{N_2O_4} + n_{NO_2}} . P$

$P_{NO_2} = \frac{100\ mol}{150\ mol} .1\ atm$

$P_{NO_2} = 0.667\ atm$

$K_P = \frac{P_{NO_2}^2}{P_{N_2O_4}}$

$K_P = \frac{(0.667)^2\ atm^2}{0.333\ \text{atm}}$

$K_P = 1.336\ atm$

$\Delta G^0 = -2.303\ RT \log K_P$

$\Delta G^0 = -2.303\ X\ 8.314\ X\ 33\ X\ \log 1.336$

$\Delta G^0 = -79.49\ J\ mol^{-1}$

68.

Solution :

Given :

$\Delta H_f^0\ (SO_2) = -297\ kJ\ mol^{-1}$

$\Delta H_f^0\ (SO_3) = -396\ kJ\ mol^{-1}$

$SO_2 + \frac{1}{2} O_2 \rightarrow SO_3 \quad \Delta H_r^0 = ?$

$\Delta H_r^0 = (\Delta H_f^0)_{compound} - \Sigma (\Delta H_f)_{elements}$

$\Delta H_r^0 = \Delta H_f^0\ (SO_3) - \left(\Delta H_f^0\ (SO_2) + \frac{1}{2} \Delta H_f^0\ (O_2) \right)$

$\Delta H_r^0 = -396\ kJ\ mol^{-1} - (-297\ kJ\ mol^{-1} + 0)$

$\Delta H_r^0 = -396\ kJ\ mol^{-1} + 297$

$\Delta H_r^0 = -99\ kJ\ mol^{-1}$

11th Chemistry Appendix Vol-1 EM.indd 286 11-01-2023 10:55:07